Question-01

"Sharjeelahmad" : key : 4

$\because C = (P+K) \bmod 26$

Plaintext	Shift +4	Cipher
s	⟶ w	
h	⟶	l
a	⟶	e
r	⟶	v
j	⟶	n
e	⟶	i
e	⟶	i
l	⟶	p
a	⟶	e
h	⟶	l
m	⟶	q
a	⟶	e
d	⟶	h

Cipher Text: "wlevniipelaeh"

with key : 4

Question-02

Step-I

Key: LAHORE

L	A	H	O	R
E	B	C	D	F
G	I/J	K	M	N
P	Q	S	T	U
V	W	X	Y	Z

STEP-II

{IA MG OI NG TO KA RA CH IW IT HS EC UR IT YX}

STEP-III : {same row: SROW, same col: SCOL Diff row & c-1 DIFF}

Pair	Rule	Cipher
IA	SCOL	QB
MG	SROW	NI
OI	DIFF	AM
NG	DIFF	GI
TO	SROW	YD
KA	DIFF	IH
RA	SROW	LH
CH	SROW	KC
IW	DIFF	QA
IT	DIFF	MQ

Pair	Rule	Cipher
HS	SCOL	CX
EC	SROW	BD
UR	DIFF	ZF
IT	DIFF	MQ
YX	SROW	ZY

Cipher: "QBNIAMGIYDIHLHKCQAMQCXBDZ FMQZY"

STEP-IV

Cipher	Decrypt	Rule
QB	IA	SCOL
NI	MG	SROW
AM	OI	DIFF
GI	NG	DIFF
YD	TO	SROW
IH	KA	DIFF
LH	RA	SROW
KC	CH	SROW
QA	IN	DIFF
MQ	IT	DIFF
CX	HS	SCOL
BD	EC	SROW
ZF	UR	DIFF
MQ	IT	DIFF
ZY	YX	SROW

Decrypted: "IAMGOINGTOKARACHIINITHSECURITYX"

Question-03

Text: "TODAYISVERYHOTDAY"

Key : SUMMER

$\therefore E_i = (P_i + K_i) \bmod 26$

Text	Ki	(P+K) mod 26	L
T	S	(19+18) % 26 = 11	L
O	U	(14+20) % 26 = 8	I
D	M	(3+12) % 26 = 15	P
A	M	(0+12) % 26 = 12	M
Y	E	(24+4) % 26 = 2	C
I	R	(8+17) % 26 = 25	Z
S	S	(18+18) % 26 = 10	K
V	U	(21+20) % 26 = 15	P
E	M	(4+12) % 26 = 16	Q
R	M	(17+12) % 26 = 3	D
Y	E	(24+4) % 26 = 2	C
H	R	(7+17) % 26 = 24	Y
O	S	(14+18) % 26 = 6	G
T	U	(19+20) % 26 = 13	N
D	M	(3+12) % 26 = 15	P
A	M	(0+12) % 26 = 12	M
Y	E	(24+4) % 26 = 2	C

Cipher : LIPMCZKPQDCYGNPMC

Decryption

Cipher	Key	(C-K) mod 26	
L	S	(11-18) mod 26 =	19 : T
I	U	(8-20) mod 26 =	14 : O
P	M	(15-12) mod 26 =	3 : D
M	M	(12-12) mod 26 =	0 : A
C	E	(2-4) mod 26 =	24 : Y
Z	R	(25-17) mod 26 =	8 : I
K	S	(10-18) mod 26 =	18 : S
P	U	(15-20) mod 26 =	21 : V
Q	M	(16-12) mod 26 =	4 : E
D	M	(3-12) mod 26 =	17 : R
C	E	(2-4) mod 26 =	24 : Y
Y	R	(24-17) mod 26 =	7 : H
G	S	(6-18) mod 26 =	14 : O
N	U	(13-20) mod 26 =	19 : T
P	M	(15-12) mod 26 =	3 : D
M	M	(12-12) mod 26 =	0 : A
C	E	(2-4) mod 26 =	24 : Y

TEXT: "TODAYISVERYHOTDAY"

Question - 04

Text = Summer

$$\text{Matrix} = \begin{bmatrix} 10 & 15 & 9 \\ 6 & 2 & 4 \\ 8 & 5 & 2 \end{bmatrix}$$

$\because C = KP \bmod 26$

"SUM MER"

$$P_1 = \begin{bmatrix} 18 \\ 20 \\ 12 \end{bmatrix}, P_2 = \begin{bmatrix} 12 \\ 4 \\ 17 \end{bmatrix}$$

$$P = \begin{bmatrix} 18 & 12 \\ 20 & 4 \\ 12 & 17 \end{bmatrix}$$

$$C = \begin{bmatrix} 10 & 15 & 9 \\ 6 & 2 & 4 \\ 8 & 5 & 2 \end{bmatrix} \begin{bmatrix} 18 & 12 \\ 20 & 4 \\ 12 & 17 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 180+300+108 & 120+60+153 \\ 108+40+48 & 72+8+68 \\ 144+100+24 & 96+20+34 \end{bmatrix}$$

$$C = \begin{bmatrix} 588 & 333 \\ 196 & 148 \\ 268 & 150 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 16 & 21 \\ 14 & 18 \\ 18 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 16 \\ 14 \\ 16 \end{bmatrix} : \begin{bmatrix} 0 \\ 0 \\ I \end{bmatrix}, \begin{bmatrix} 21 \\ 0 \\ 20 \end{bmatrix} : \begin{bmatrix} V \\ S \\ U \end{bmatrix}$$

Cipher ⇒ ~~QOIVSU~~ QOIVSU

Decryption

$$\therefore P = K^{-1}C \bmod 26$$

Check if matrix invertible

$$\det(K) = 10[(2\times2) - (4\times5)] + 5[(6\times2) - (4\times8)] + 9[(6\times5) - (8\times2)]$$

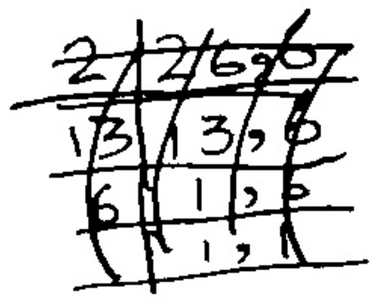
$$\Rightarrow -160 + 300 + 126$$

$$\det(K) = 266$$

$$\Rightarrow 266 \bmod 26 = 6$$

Take GCD of 26 and 6

$$\gcd(6, 26) = 2$$



As, GCD is not equeal to 1 as hill climb

26: 1, 2, 13, 26
6: 1, 2, 3, 6

decryption requires → This cipher decryption is not possible.